Do Self-Supervised Vision Models Learn What Experts See?

Attention Alignment with Human-Annotated Architectural Features

ISY5004 Intelligent Sensing Systems Practice Project

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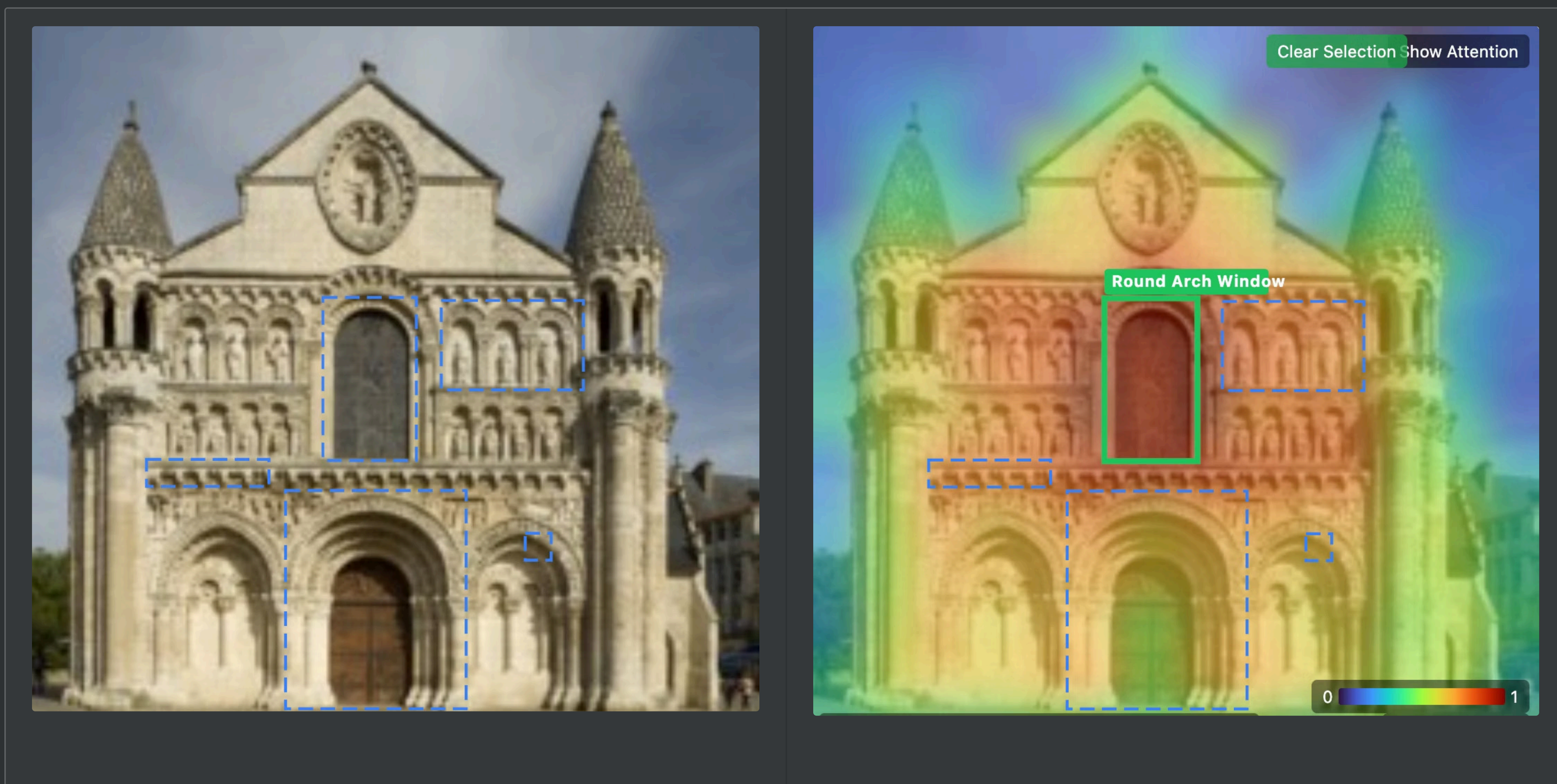
Motivation & Research Gap

Self-supervised learning (SSL) models learn visual representations without labels and achieve strong performance on downstream tasks. However, a fundamental question remains: **Do these models attend to the same visual features that human experts consider diagnostic, or do they exploit statistical shortcuts invisible to humans?**

As a corollary, **does fine-tuning shift attention toward expert-identified features, or do models discover alternative discriminative regions?**



The Gap We Address

Existing self-supervised learning (SSL) benchmarks focus on classification accuracy, neglecting **which** image regions drive predictions. This is crucial for trust and deployment: does a model genuinely recognise architectural features, or does it exploit statistical shortcuts?

Limitations of Existing Tools

While tools like BertViz visualise attention and Captum measures faithfulness, they don't quantitatively align model attention with human expert diagnoses.

Current work in medical imaging is a start, but lacks a comprehensive comparison across multiple SSL paradigms or an analysis of attention shifts during fine-tuning.

Our Solution

We introduce a multi-metric quantitative benchmark to measure alignment between SSL attention patterns and expert-annotated *"characteristic architectural features."*

Comprehensive Analysis

Our approach uses both threshold-dependent (IoU) and threshold-free (MSE, KL divergence, EMD) metrics, comparing multiple SSL paradigms and analysing how fine-tuning impacts attention alignment.

Dataset: WikiChurches

Proposed by Barz & Denzler (NeurIPS 2021), this dataset evaluates architectural feature alignment.
It contains 631 bounding box annotations of characteristic visual features for 139 churches from four major categories

9,485 Images
Full Dataset Size

4,588 Images
Used for Fine-Tuning

631 Bounding Boxes
Across Different Features

139 Expert Annotations
For Attention Evaluation

Figure 1: Example images from the 14 main styles (≥ 20 images) in the WikiChurches dataset. The numbers specify the number of images in the respective category, including sub-categories.

35th Conference on Neural Information Processing Systems (NeurIPS 2021) Track on Datasets and Benchmarks.

RESEARCH AGENDA

Research Questions and Approaches

Q1: Expert-Attention Alignment

Do vision models attend to the same diagnostic features identified by human experts? We quantify this alignment using metrics like Intersection over Union (IoU), Mean Squared Error (MSE), KL Divergence, and Earth Mover's Distance (EMD) across 7 diverse models and their 12 layers.

Tools: Attention heatmap overlays, multi-metric dashboard, model leaderboard.

Q2: Fine-Tuning Attention Shifts

Does fine-tuning shift model attention towards expert-identified features, and how do different strategies (Linear Probe, LoRA, Full Fine-tuning) impact this? We compare Δ IoU (fine-tuned vs. frozen) and classify outcomes as "Preserve, Enhance, or Destroy" using statistical tests.

Tools: Frozen vs. fine-tuned comparison view, attention shift visualisation, Δ IoU bar charts by method.

Q3: Attention Head Specialisation

Do individual attention heads within these models specialise in different architectural features? We compute per-head IoU for all 12 attention heads to identify those that consistently show the highest alignment with expert annotations.

Tools: Per-head attention selector, head IoU heatmap, head specialisation dashboard.

Professor's Feedback

Four suggestions that shaped our methodology

Continuous Metrics

Feedback: "Apply Gaussian filtering to bounding boxes and compare with MSE, like crowd density estimation"

Implementation: MSE, KL Divergence, EMD added alongside IoU

IMPLEMENTED ✓

Fine-Tuning Impact

Feedback: "Study whether SFT preserves, enhances, or destroys attention consistency"

Implementation: Preserve / Enhance / Destroy taxonomy applied to 6 models × 3 strategies

IMPLEMENTED ✓

Cross-Layer Aggregation

Feedback: "Aggregate attention from different layers (e.g. max pooling) to construct a new map"

Implementation: Max pooling, depth-weighted mean, ALTI

PLANNED

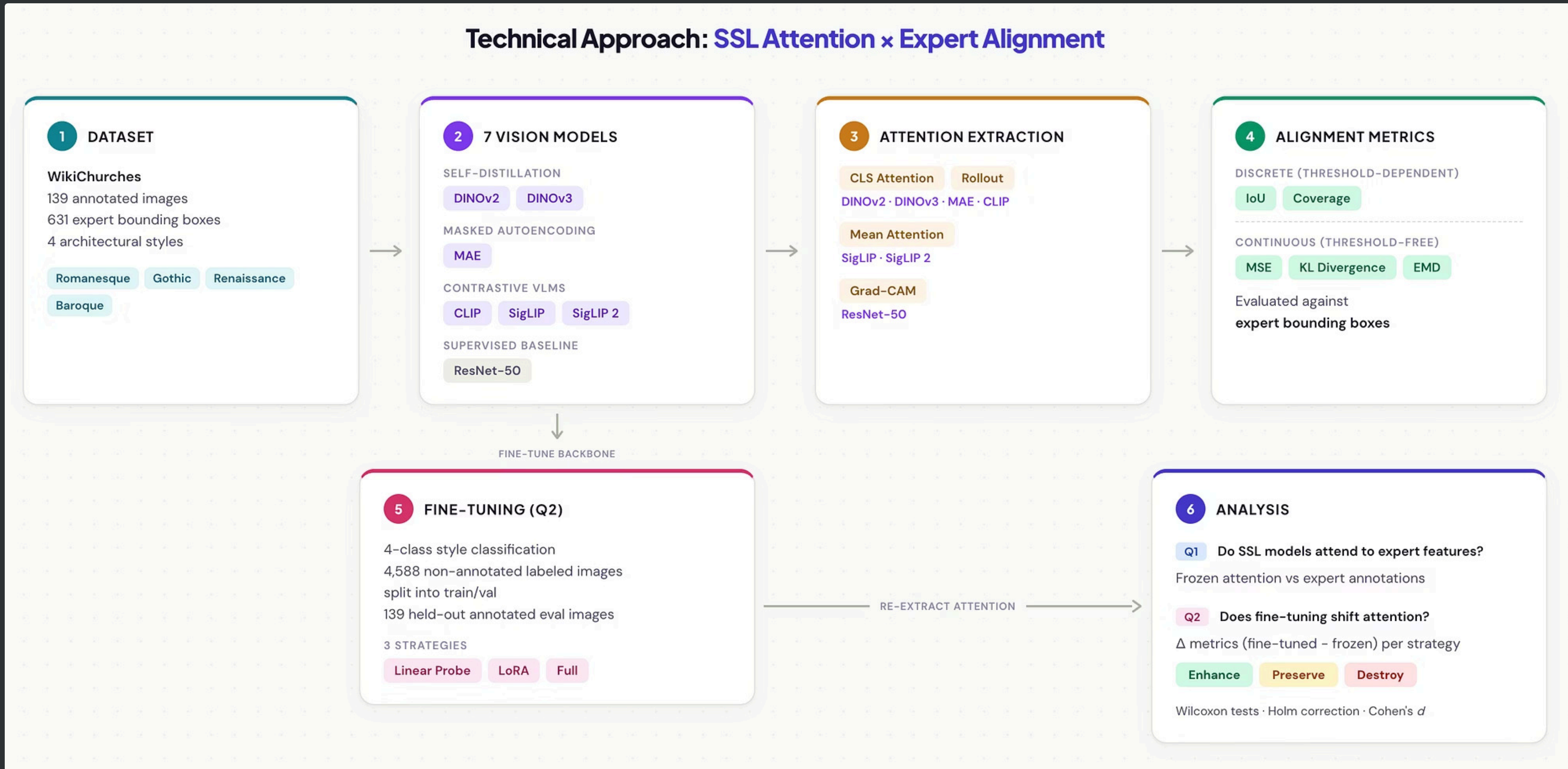
Unimodal vs Multimodal Leaderboard

Feedback: "Compare unimodal models (DINO) and VLM vision encoders (CLIP, SigLIP)"

Implementation: Paradigm grouping applied in analysis; formal dashboard split planned

PARTIAL

Architecture



7 Models Across SSL Paradigms + Baseline

All ViT-B architecture (12 layers, 768 dim, 12 heads, ~86-93M params) except ResNet-50

Model	Paradigm	Patch (px)	Method	Key Feature
DINOv2	Self-distillation	14	CLS, Rollout	4 register tokens
DINOv3	Self-distillation + Gram Anchoring	16	CLS, Rollout	RoPE encoding
MAE	Masked Autoencoding	16	CLS, Rollout	Pixel reconstruction
CLIP	Contrastive (softmax)	16	CLS, Rollout	Language-image align
SigLIP	Contrastive (sigmoid)	16	Mean	No CLS token
SigLIP 2	Contrastive (sigmoid)	16	Mean	Dense features
ResNet-50	Supervised (ImageNet)	—	Grad-CAM	CNN baseline

Attention Extraction Methods

To understand what these models learn, we employ various techniques to extract and visualize their attention patterns. These methods help us interpret how models process visual information.

CLS Token Attention

- Leverages **special class token in ViTs** to represent global image understanding, often reflecting the most salient regions.

Attention Rollout

- Aggregates attention weights across multiple transformer layers
- Provides a more comprehensive and hierarchical view of feature importance

Mean Attention

- Calculates average attention weight across all attention heads and layers for each pixel
- Offers a general overview of model focus

Grad-CAM

- Utilizes gradients of a target class flowing into the final **convolutional layer** to produce a localization map highlighting important regions for prediction.

Metrics Used to Measure Attention

To quantitatively assess how well model attention aligns with expert understanding, we define five complementary metrics across two ground truth types. Each metric captures a different aspect of alignment; together they prevent any single metric from misleading.

Binary Ground Truth Metrics

Compare attention against the **hard bounding box mask** (binary: inside or outside the box).

IoU

Spatial overlap between top-k% attention pixels and expert bounding boxes.

Higher = better.

Coverage

Fraction of total attention energy inside expert bounding boxes. No threshold needed.

Higher = better.

Soft Gaussian Ground Truth Metrics

Compare attention against a **Gaussian heatmap** derived from bounding boxes (continuous: smooth falloff from centre).

MSE

Pointwise intensity error. How closely attention intensity matches expert expectations.

Lower = better.

KL Divergence

Distribution mismatch. Penalises attention in areas where experts assign none.

Lower = better.

EMD

Earth Mover's Distance. Spatial transport cost; distinguishes near-misses from far-misses.

Lower = better.

Q1 EXPERT-ATTENTION ALIGNMENT

Frozen Model Leaderboard

Average IoU @ 90th percentile

Self-distillation > Supervised > Reconstruction > Multimodal contrastive.

These rankings are calculated using the default attention methods tagged to each model averaged over all the annotated images.

Current Ranking

1. DINOv3 — 0.133
2. ResNet-50 — 0.090
3. DINOv2 — 0.082
4. MAE — 0.070
5. CLIP — 0.049
6. SigLIP — 0.047
7. SigLIP2 — 0.047

Self-distillation leads the frozen benchmark:

DINOv3 remains the strongest frozen model on best-layer IoU.

The supervised baseline beats all multimodal contrastive models in the frozen setting.

The contrastive VLMs still cluster tightly, with CLIP slightly above the SigLIP pair.

Q2 FINE-TUNING ATTENTION SHIFTS

Fine-Tuning Experiment Design

Addressing Feedback #2: does SFT preserve, enhance, or destroy consistency?

Linear Probe:

~3K params

Classifier head only

Backbone frozen

Baseline — attention unchanged

LoRA:

~300K params

Low-rank adapters on attention layers + head

Parameter-efficient; preserves pre-training

Full Fine-tune:

86~93M params

Entire backbone + classifier head

Maximum adaptation capacity

Training Configuration:

4-class style classification (Romanesque, Gothic, Renaissance, Baroque)

- 3 epochs, batch size 16
- Cosine LR + warmup
- Class-weighted loss
- 139 annotated images strictly excluded from training

Measurement:

Δ IoU = IoU(fine-tuned) − IoU(frozen), per image, paired Wilcoxon tests with Holm correction

Taxonomy:

- **Enhance** ($\Delta > 0$, significant)
- **Preserve** ($\Delta \approx 0$, not significant)
- **Destroy** ($\Delta < 0$, significant)

> Checkpoint selection: Best classification validation accuracy on validation split.

Q2 FINE-TUNING ATTENTION SHIFTS

Post Fine-Tuning Results Analysis (IoU@90)

Enhance:

Contrastive models (CLIP, SigLIP)

- Full ΔIoU@90: CLIP > SigLIP/2 > MAE
- LoRA also improves CLIP, SigLIP, and SigLIP2, but not as much as Full.

Preserve:

Self-distillation & reconstruction

- DINOv3 LoRA: ΔIoU +0.007
- Near-flat ΔIoU@90 for DINOv2 and DINOv3 with other strategies
- Baseline sanity check with Linear Probe: Δ ≈ 0 across all models

Destroy:

Directionally concerning but not catastrophic

- DINOv2 Full: Negative ΔIoU
- DINOv3 Full: ΔIoU near zero at p90 but worsens some threshold-free metrics.

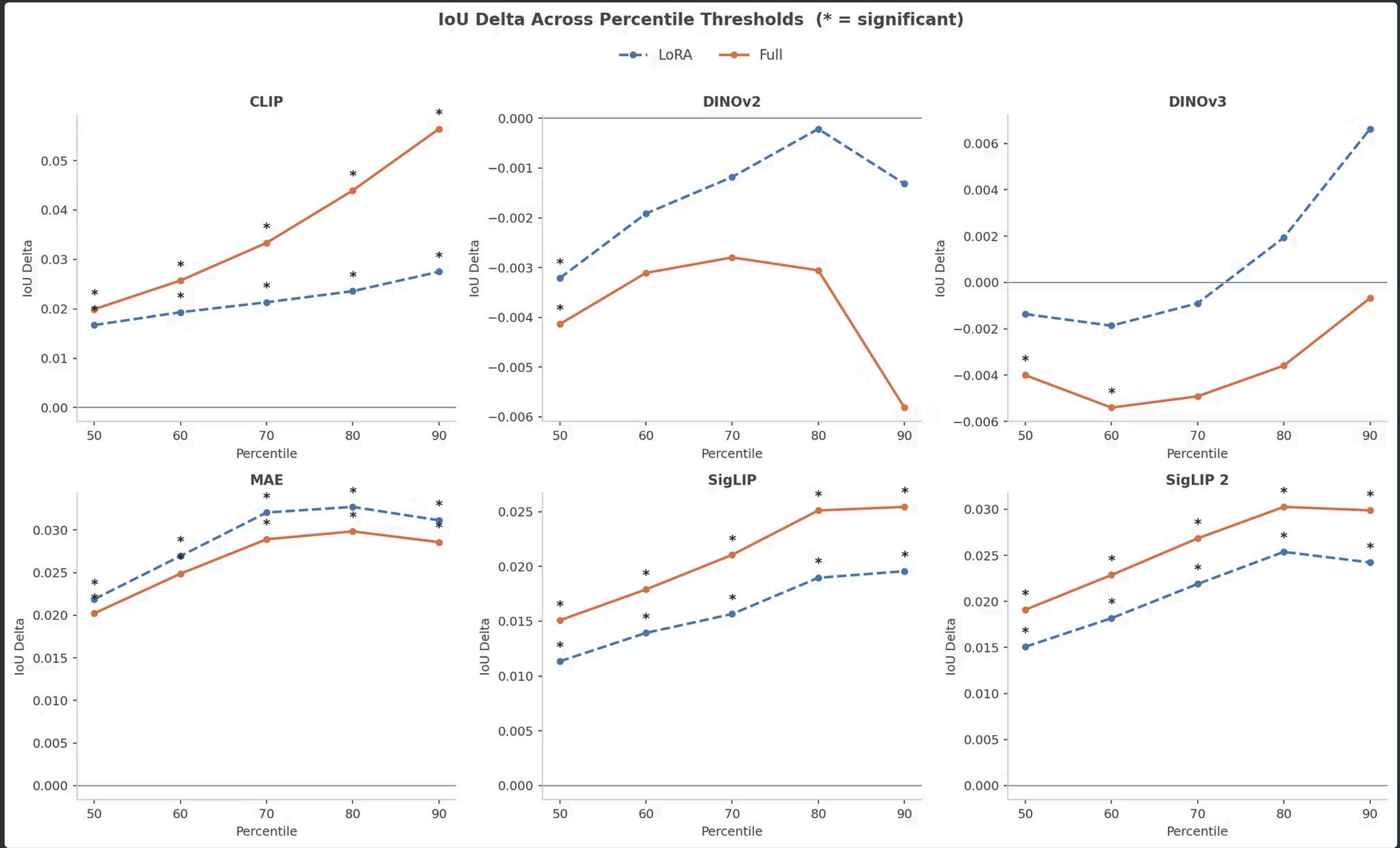
Key Findings:

- Full > LoRA > Linear Probe for Contrastive Models (CLIP, SigLIP, and SigLIP2)
- MAE is an exception where both LoRA and Full improve strongly, with LoRA slightly ahead at IoU@90.
- Fine-tuning appears to worsen Self-distillation models (DinoV2, DinoV3)

Q2

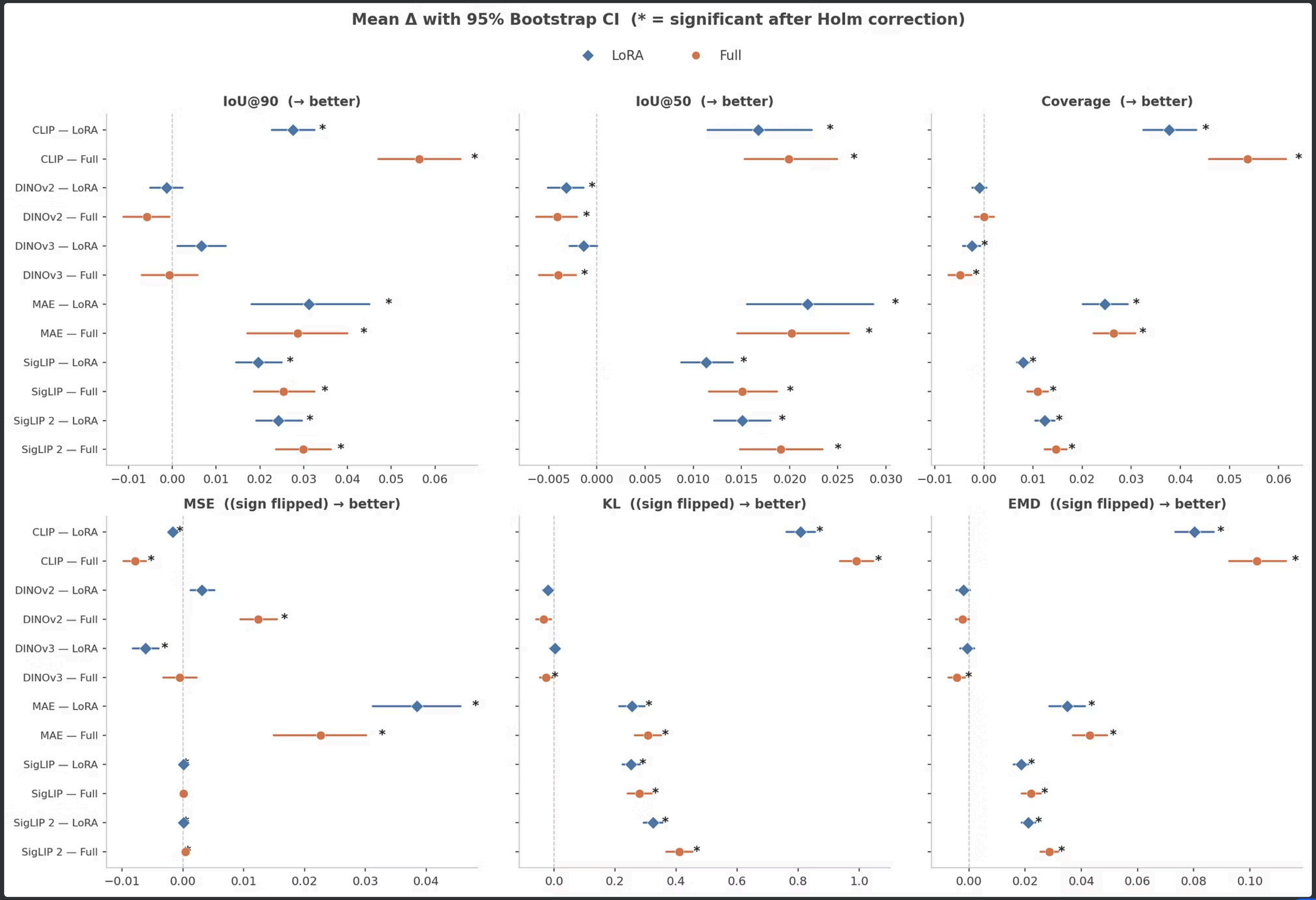
FINE-TUNING ATTENTION SHIFTS

Results: ΔIoU Across Fine-Tuned Models



Made with GAMMA

Results: ΔMetrics Across Fine-Tuned Models



Q2 FINE-TUNING ATTENTION SHIFTS

Key Learning Outcomes

Targeted Improvements

- Fine-tuning helps the contrastive models most: CLIP, SigLIP, and SigLIP2.

- CLIP shows the clearest gain, with strong positive IoU shifts across the main percentiles.

Limited Impact on DINO

- DINOv2 and DINOv3 change little under fine-tuning, and can even dip slightly.

- DINOv3 starts from the strongest frozen alignment, leaving less room to improve.

LoRA vs. Full Fine-tuning

- Full fine-tuning gives the biggest gains; LoRA is usually the best lighter-weight alternative.

- MAE also improves under both LoRA and Full.

Preliminary Insights: No single strategy wins everywhere. Fine-tuning effects depend strongly on the pre-training objective.

DEMO :)

Conclusion & Next Steps

Current Status

Complete infrastructure: dataset pipeline, model wrappers, attention extraction, metric computation, fine-tuning workflow, and interactive web application all implemented and producing results

Next Steps

- Deeper analysis of fine-tuning results
- Implement cross-layer aggregation
- Complete Q3 per-head specialization

Stretch Goals

- Expand to other architectural feature types
- Explore attention head specialization patterns

Key Related Work & Our Extensions

Paper	What It Covers	How This Project Extends the Work
DINO (Caron et al., 2021)	Self-supervised ViTs learn features containing semantic segmentation info not present in supervised models.	Qualitative observation quantified: attention alignment with 631 expert-annotated architectural features measured via IoU across 7 models, revealing emergent segmentation varies by training objective.
Teaching Matters (Walmer et al., 2023)	Systematic comparison showing ViTs learn different attention behaviors depending on supervision paradigm.	Extends from characterizing attention *patterns* to measuring attention *alignment* with external domain expertise, providing a quantitative answer on whether patterns correspond to expert-relevant features.
Park et al. (2023)	Contrastive SSL captures global/shape patterns while masked image modeling focuses on local textures.	Global-vs-local distinction externally validated: contrastive (CLIP, SigLIP) and self-distillation (DINO) models show different IoU profiles against architectural features, providing expert-grounded evidence.
Toker et al. (2025)	Eye-tracking study showing DINO-trained ViTs mimic human visual attention patterns.	Extends from human gaze as ground truth to domain expert annotations, and from single-model to cross-paradigm comparison of 7 models, testing if "human-like attention" generalizes to specialized expert knowledge.
WikiChurches (Barz & Denzler, 2021)	Fine-grained dataset of 9,485 church images with style labels and 631 expert bounding box annotations.	Bounding boxes, originally for recognition evaluation, repurposed as ground truth for attention alignment—a novel use case transforming object-detection annotations into an interpretability benchmark.
Grad-CAM (Selvaraju et al., 2017)	Gradient-based saliency maps for CNNs highlighting important image regions for model decisions.	Grad-CAM applied to ResNet-50 as a CNN baseline within a unified evaluation framework, measuring CNN saliency and ViT attention maps against the same 631 expert annotations using identical metrics.
Attention Rollout (Abnar & Zuidema, 2020)	Computes token-level input attribution by recursively combining attention matrices across transformer layers.	Rollout implemented across 4 ViT architectures spanning 3 training paradigms and evaluated against expert bounding boxes—moving from a general-purpose attribution tool to a paradigm-comparative evaluation instrument.
Vision Transformers Need Registers (Darcet et al., 2024)	Identifies high-norm "artifact" tokens in ViT attention and proposes learnable register tokens as a solution.	Register tokens explicitly handled in attention extraction for DINOv2/DINOv3; effect on expert-aligned attention implicitly evaluated by comparing register-equipped models against register-free architectures.
Chefer et al. (2021)	Class-specific relevance maps via gradient-based propagation; evaluates ViT explanations using IoU and Pointing Game against segmentation masks.	Evaluation shifts from model-internal faithfulness to expert-grounded alignment, with paired statistical testing across 7 models.
Attention IoU (Kashefi & Barekatain, 2025)	Formalizes IoU-based metric for evaluating attention map alignment, applied to bias detection in facial attributes.	Extends from bias detection to domain-specific expert annotations; adds paired Wilcoxon tests with Holm-Bonferroni correction, and introduces Δ IoU metric to quantify fine-tuning's shift in attention alignment.
Demsar (2006)	Recommends Wilcoxon signed-rank tests with Holm-Bonferroni correction for rigorous multi-classifier comparison.	Recommended statistical framework applied to attention alignment rather than classification accuracy, yielding a novel three-outcome taxonomy (Preserve / Enhance / Destroy) for characterizing fine-tuning effects on attention.

Thank you for your attention!

Project Highlights

Beyond Classification

Measures whether models attend to the right architectural regions, not just whether they classify correctly

Expert Ground Truth

Uses human expert bounding boxes as reference, not weak heuristics or proxy labels

7 Model Variants

Compares DINOv2, DINOv3, MAE, CLIP, SigLIP, SigLIP2, and ResNet50 across training paradigms

Frozen & Fine-Tuned

Evaluates both pretrained weights and after fine-tuning on style recognition

Interactive Tool

Delivers a complete analysis platform, not just static experiment results

Experimental Design

We conducted a comprehensive evaluation of model attention across diverse architectures and fine-tuning strategies.

Frozen Model Benchmark

- 7 model variants: DINOv2, DINOv3, MAE, CLIP, SigLIP, SigLIP2, ResNet50
- Systematic layer sweep for each architecture
- Method-aware evaluation: CLS, rollout, mean attention, and Grad-CAM

Fine-Tuning Strategy

3 methods: LoRA, Linear Probe, Full Fine-tuning

Strategy-Aware Evaluation

We compare frozen vs fine-tuned attention after selecting one checkpoint per model × strategy by best classification validation accuracy on a shared non-annotated validation split.

Held-Out Attention Analysis

Attention alignment is evaluated only on the untouched 139 expert-annotated images, using IoU, Coverage, MSE, KL, and EMD to measure whether fine-tuning shifts focus toward expert-defined architectural regions.

Goal

To analyse the model's ability to **focus** on the **diagnostically relevant architectural features identified by human experts** when performing **style classification**.

Q1: Frozen Model Leaderboard

Addressing Feedback #4: unimodal vs multimodal comparison

Paradigm Ranking:

Unimodal self-distillation leads:

DINOv3 (0.133) achieves 1.6× the IoU of DINOv2 (0.082). Self-distillation learns part-level segmentation.

Supervised > Multimodal:

ResNet-50 (0.090) outperforms all VLMs. Language alignment doesn't help localisation.

VLMs cluster together:

CLIP/SigLIP/SigLIP2 all ≈ 0.047~0.049. Contrastive objective produces similar attention patterns.

Reconstruction is worst:

MAE (0.037) — pixel reconstruction ≠ object localisation.

Note: We'll add the bar chart visualization later showing Frozen IoU @ 90th Percentile (Best Layer).

Comparison: ViTs uses attention-derived maps; ResNet uses Grad-CAM as a “attention-like localisation map"

Q2 FINE-TUNING ATTENTION SHIFTS

Post Fine-Tuning Results Analysis (IoU@90)

Enhance:

Contrastive models (CLIP, SigLIP)

- CLIP Full: ΔIoU@90 = +0.056 = 0.075 - 0.018
- SigLIP/2 Full: Δ IoU@90 =+0.025~0.030
- MAE LoRA/Full: Δ IoU@90 =+0.029~0.031
- LoRA also improves CLIP, SigLIP, and SigLIP2, but not as much as Full.

Preserve:

Self-distillation & reconstruction

- DINOv3 LoRA: ΔIoU +0.007 (not significant)
- Near-flat ΔIoU@90 for DINOv2 and DINOv3 with other strategies
- Linear Probe: Δ ≈ 0 across all models (Baseline sanity check)

Destroy:

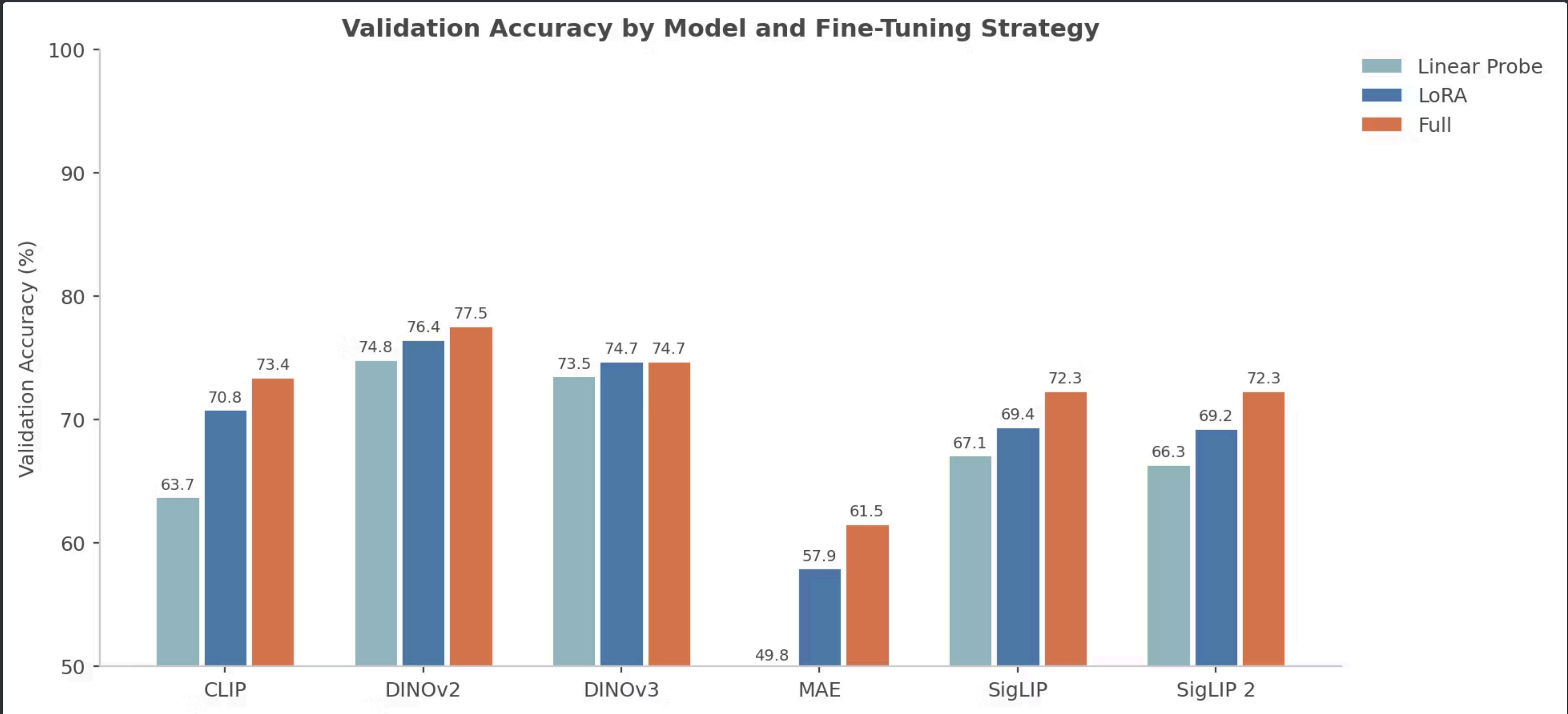
Directionally concerning but not catastrophic

- DINOv2 Full: ΔIoU slightly negative at p90 and more negative at lower IoU percentiles.
- DINOv3 Full: ΔIoU near zero at p90 but worsens some threshold-free metrics.

Key Findings:

- Full > LoRA > Linear Probe for Contrastive Models (CLIP, SigLIP, and SigLIP2)
- MAE is an exception where both LoRA and Full improve strongly, with LoRA slightly ahead at IoU@90.
- Fine-tuning appears to worsen Self-distillation models (DinoV2, DinoV3)

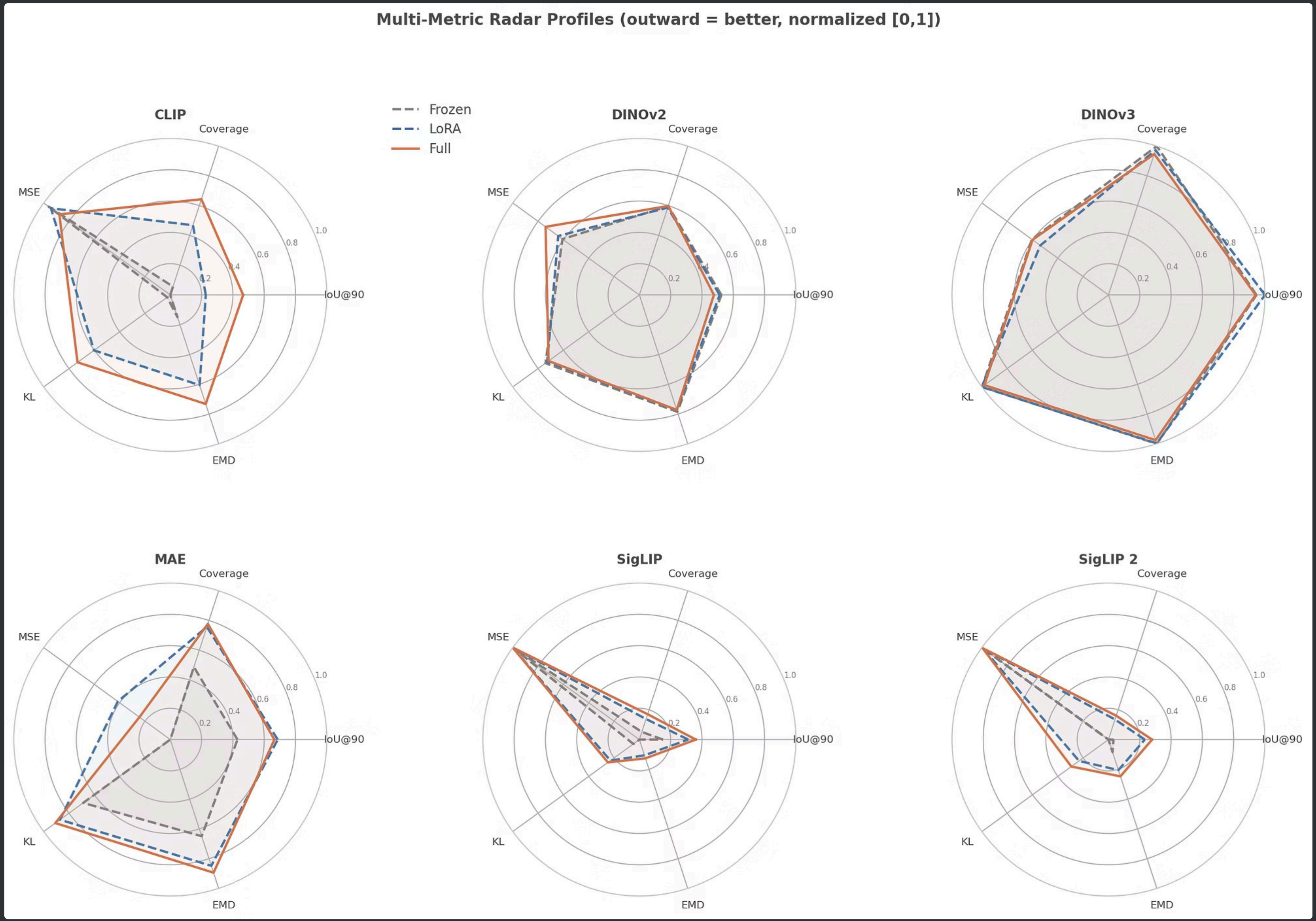
Results Visualization: Accuracy Uplift



Q2

FINE-TUNING ATTENTION SHIFTS

Results Visualization – Radar Profiles



Deep Dive: Fine-Tuning Outcomes & Model Behaviour

Targeted Improvements

- Fine-tuning most strongly improves the contrastive models: CLIP, SigLIP, and SigLIP2.

- CLIP shows the largest attention shift, with significant positive IoU deltas from p90 through p50.

- SigLIP and SigLIP2 also improve consistently across IoU percentiles, with Full slightly above LoRA.

- MAE also improves broadly after the deterministic analysis fix, rather than staying flat.

Limited Impact on DINO

- DINOv2 and DINOv3 mostly preserve their attention patterns under fine-tuning, and can even worsen slightly.

- DINOv3 starts with the strongest frozen alignment, which leaves relatively little headroom to improve.

- DINOv2 Full is slightly negative across IoU percentiles; DINOv3 Full is near-flat at p90 and negative at lower percentiles.

LoRA vs. Full Fine-tuning

- Linear Probe leaves attention essentially unchanged, validating the frozen-backbone control.

- Full fine-tuning gives the largest gains on the contrastive models; LoRA is a useful second-best compromise.

- MAE is another positive case: both LoRA and Full improve IoU, Coverage, MSE, KL, and EMD in the corrected primary run.

- Continuous metrics add nuance:
  - CLIP improves KL and EMD even though MSE worsens.
  - DINOv3 is mostly flat on IoU@90 but worsens some threshold-free metrics.

Conclusion: This demonstrates that one strategy does not win everywhere. The required strategy effects depend strongly on the pre-training objective.

Key Takeaways

Summary of findings and next steps

Feedback #1 — Continuous Metrics

5 complementary metrics implemented (IoU + Coverage + MSE + KL + EMD). Threshold-free metrics reveal patterns IoU alone misses.

Feedback #2 — Preserve / Enhance / Destroy

Contrastive models Enhance; self-distillation Preserves; no model Destroys. Full fine-tuning returns the highest gains using IoU as evaluation.

Feedback #4 — Unimodal vs Multimodal

Unimodal self-distillation > supervised > multimodal.

Language alignment does not improve localisation of architectural features.

But lower frozen alignment does not mean lower plasticity: CLIP and the SigLIP models improve the most after fine-tuning.

Key Insight:

Frozen IoU vs Δ IoU: pre-training determines alignment & how much attention can move on finetuning.

DINO starts high and moves little; contrastive models start lower but adapt much more.

Next Steps:

Feedback #3 (cross-layer aggregation) + Q3 (per-head specialisation) + feedback #4 (formal dashboard split, SAM).

Measuring Attention-Expert Alignment

Addressing Feedback #1: continuous metrics beyond IoU

Metric	Measures	Threshold?
IoU	Spatial overlap of top-k attention with expert regions	Yes
Coverage	Fraction of attention energy inside expert regions	No
Gaussian MSE	Distance from attention to Gaussian-blurred GT	No
KL Divergence	Distribution divergence vs ground truth	No
EMD	Optimal transport cost between distributions	No

Statistical Rigor

- Paired Wilcoxon signed-rank tests (139 image pairs per comparison)
- Holm-Bonferroni correction
- Cohen's d effect sizes with 95% bootstrap CIs

Interactive Visualization Platform

Precompute → HDF5/SQLite/PNG cache → FastAPI (30 endpoints, 4 routers) → React + Vite

Gallery

Browse all 139 annotated images with style labels and attention overlays

Dashboard

Model leaderboard, layer progression charts, metric comparisons across paradigms

Image Detail

Single image analysis with attention heatmaps, expert annotations, layer-wise metrics

Compare

Side-by-side model comparison with frozen vs fine-tuned attention patterns

See: Demo

Deliverables & Progress

Codebase

Reproducible Python pipeline, model wrappers, and analysis utilities.

Web App

Interactive dashboard, model comparison tools, and image gallery.

Benchmark Results

Comprehensive leaderboard comparing 7 frozen and 18 fine-tuned models.

Documentation

Analysis of Q2 performance, feature cache, and metrics database.

Remaining Work & Roadmap

Feedback #3: Cross-Layer Aggregation

- Max pooling across 12 layers: Pixel-wise max → comprehensive saliency map
- Depth-weighted mean pooling: Exponential decay weighting deeper (semantic) layers
- ALTI (Aggregation of Layer-wise Token Interactions): State-of-the-art recursive aggregation (Ferrando et al., 2022)

Q3: Per-Head Specialization

- Per-head IoU × feature-type matrix
- Head ranking by alignment consistency
- Voita et al. (2019), Caron et al. (2021)

Feedback #4: Paradigm-Split Leaderboard

- Formal unimodal vs multimodal grouping in dashboard
- Sub-group averages for IoU, MSE, Coverage
- Potentially add SAM (Segment Anything) encoder
- Text-prompted evaluation for VLMs (CLIP, SigLIP)

Other Open Items

- Attention shift diff-map visualization
- Fine-tuning training loop test coverage
- Feature-specific forgetting analysis (per architectural element)